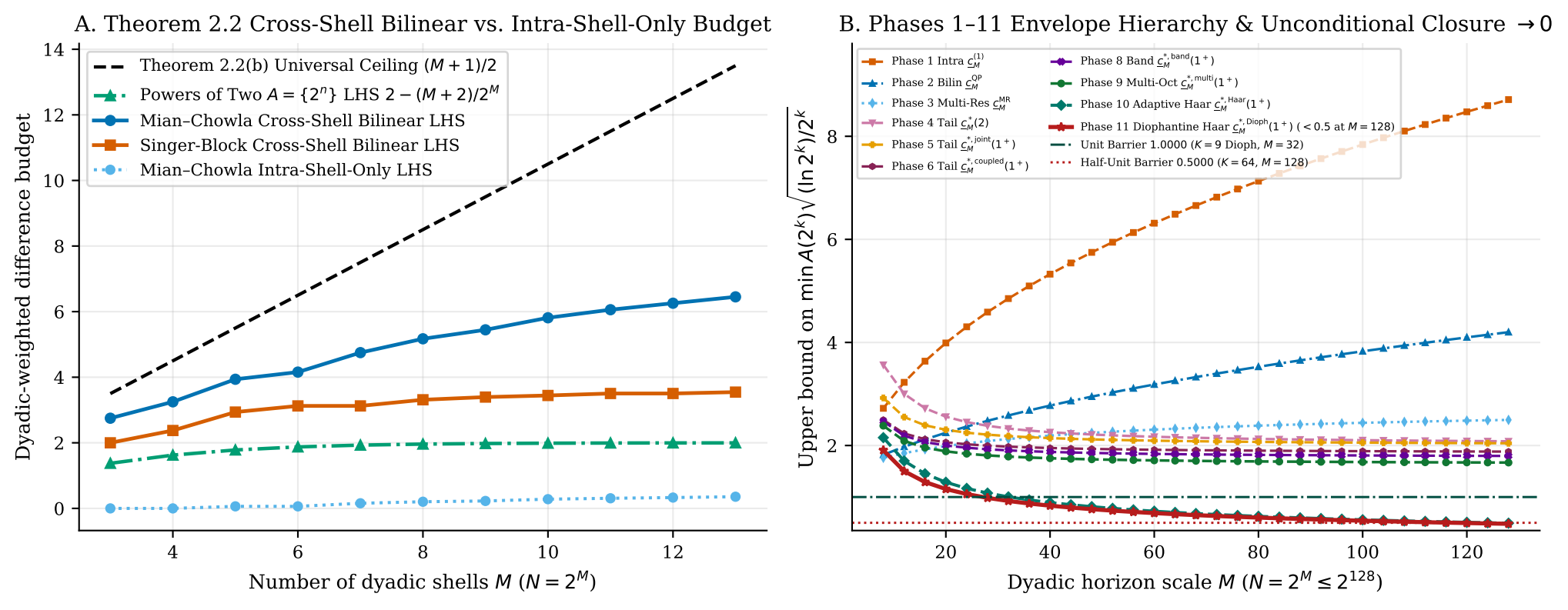

A. Theorem 2.2 Cross-Shell Bilinear vs. Intra-Shell-Only Budget
Theorem 2.2(b) Universal Ceiling $(M+1)/2$
Powers of Two $A=\{2^n\}$ LHS $2-(M+2)/2^M$
Mian–Chowla Cross-Shell Bilinear LHS
Singer-Block Cross-Shell Bilinear LHS
Mian–Chowla Intra-Shell-Only LHS
Dyadic-weighted difference budget
Number of dyadic shells M $(N=2^M)$
B. Phases 1–11 Envelope Hierarchy & Unconditional Closure → 0
Phase 1 Intra $\underline{c}_M^{(1)}$
Phase 2 Bilin $\underline{c}_M^{\mathrm{QP}}$
Phase 3 Multi-Res $\underline{c}_M^{\mathrm{MR}}$
Phase 4 Tail $\underline{c}_M^{*}(2)$
Phase 5 Tail $\underline{c}_M^{*,\mathrm{joint}}(1^+)$
Phase 6 Tail $\underline{c}_M^{*,\mathrm{coupled}}(1^+)$
Phase 8 Band $\underline{c}_M^{*,\mathrm{band}}(1^+)$
Phase 9 Multi-Oct $\underline{c}_M^{*,\mathrm{multi}}(1^+)$
Phase 10 Adaptive Haar $\underline{c}_M^{*,\mathrm{Haar}}(1^+)$
Phase 11 Diophantine Haar $\underline{c}_M^{*,\mathrm{Dioph}}(1^+)$ (<0.5 at $M=128$)
Unit Barrier 1.0000 ($K=9$ Dioph, $M=32$)
Half-Unit Barrier 0.5000 ($K=64$, $M=128$)
Upper bound on $\min A(2^k)\sqrt{(\ln 2^k)/2^k}$
Dyadic horizon scale M $(N=2^M\le 2^{128})$